Infinite NPA depth, finite semidefinite lift

An exact quantum region in the simplest Bell experiment

Alice and Bob choose between two measurements with outcomes ± 1. Write $a_x = \langle A_x \rangle$, $b_y = \langle B_y \rangle$ and $E_{xy} = \langle A_x B_y \rangle$. Retain

$$M = a_0 + b_0, \qquad S = E_{00} + E_{01} + E_{10} - E_{11}. \tag{1}$$

Their quantum region $\mathcal{W} = \{(M, S) : p \in \mathcal{Q}_{222}\}$ includes all quantum behaviors, with no equality imposed on the marginal averages. Our preceding Letter [1] showed that every finite NPA level misses supporting lines of this region. We now give one exact finite semidefinite lift of the same region. If $\mathcal{W}_L$ is its level-L NPA outer approximation, then

$$\boxed{\begin{array}{c} \mathcal{W} \subsetneq \mathcal{W}_L \quad \text{for every finite } L, \\ \operatorname{sxdeg}(\mathcal{W}) = 2. \end{array}} \tag{2}$$

We construct one lift with positive semidefinite (PSD) blocks of orders five and four. Extension degree is the smallest possible largest block order among all finite lifts; its value two follows separately from a planar representation theorem.

Why one angle suffices. To determine $\mathcal{W}$, consider an arbitrary supporting inequality $hS + \alpha M \leq \lambda$, with real h, α. Jordan's lemma reduces its validity to positivity of two-qubit slacks $\lambda I - H$, where

$$H = h\,\mathrm{CHSH} + \alpha(ZI + IZ),$$
$$A_1 = cZ + \sqrt{1-c^2}X, \quad B_1 = dZ + \sqrt{1-d^2}X.$$

Here $A_0 = B_0 = Z$ and $c, d \in [-1, 1]$ are the cosines of the local measurement angles. Deterministic strategies attain $\ell = 2|h| + 2|\alpha|$, so $\lambda \geq \ell$ is necessary. Above this bound, the first three elementary symmetric functions of the eigenvalues of $\lambda I - H$ are nonnegative. Its determinant therefore decides positivity: if it too is nonnegative, the characteristic polynomial of $H - \lambda I$ has nonnegative coefficients and hence no positive root.

Put $x = (1-c)/2$, $y = (1-d)/2$, and write $D(x, y) = \det(\lambda I - H)$. The determinant takes the form

$$D(x, y) = F(xy) + 64h^2(4h^2 - \alpha^2)xy(x + y), \tag{3}$$

where F depends on the product alone. In the nonclassical range $|\alpha| \leq |h|$, the coefficient of $x + y$ is nonnegative. At fixed product, $x + y \geq 2\sqrt{xy}$, with equality at $x = y = \sqrt{xy}$. Thus equal overlaps minimize the determinant; positivity there implies positivity for every pair. For $|\alpha| \geq |h|$, the quantum support is the deterministic value ℓ: average the two signed one-sided tilted-CHSH bounds and add the remaining marginal slack. Equal overlaps attain this value too. Hence they suffice for *every* support direction.

At equal overlaps $c = d$, the singlet decouples. In the remaining three-dimensional block, set $c = (1-t^2)/(1+t^2)$ and multiply the slack by $(1+t^2)^2$. With

$$p_0 = (1+t^2)^2, \qquad p_1 = 1 - t^4, \qquad p_2 = 1 + 4t^2 - t^4,$$

the result is the quartic matrix

$$K(t) = \begin{pmatrix} \lambda p_0 - 2hp_1 & -2\alpha p_0 & 0 \\ -2\alpha p_0 & \lambda p_0 - 2hp_2 & -8ht^3 \\ 0 & -8ht^3 & \lambda p_0 + 2hp_2 \end{pmatrix}. \tag{4}$$

Its value at zero and leading coefficient force $\lambda \geq \ell$, which also controls the singlet. The missing endpoint follows by continuity. We have therefore proved

$$\max_{\mathcal{W}} (hS + \alpha M) \leq \lambda \quad \Longleftrightarrow \quad K(t) \succeq 0 \quad \forall t \in \mathbb{R}. \tag{5}$$

All three support coefficients still enter *affinely*. This is what will give one lift for the whole region.

From one variable to one finite lift. A PSD real univariate matrix polynomial is a polynomial matrix sum of squares, including on its singular boundary [2]. For a quartic, the factors have degree at most two. For $\xi \in \mathbb{R}^3$, the parity of Eq. (4) splits the nine quadratic features into

$$Z_+ = (\xi_1, \xi_2, t\xi_3, t^2\xi_1, t^2\xi_2)^T,$$
$$Z_- = (\xi_3, t\xi_1, t\xi_2, t^2\xi_3)^T.$$

Consequently Eq. (5) is equivalent to the polynomial identity

$$\begin{array}{c} \xi^T K(t) \xi = Z_+^T G Z_+ + Z_-^T J Z_-, \\ G \succeq 0, \quad J \succeq 0, \end{array} \tag{6}$$

with G of order five and J of order four. Matching coefficients gives sixteen affine equations.

Dualizing gives the membership lift

$$\begin{array}{l} \mathcal{W} = \{(m(z), s(z)) : z \in \mathbb{R}^{16}, \\ \qquad \ell(z) = 1,\ U(z), V(z) \succeq 0\}. \end{array} \tag{7}$$

The matrices U, V are the adjoints of the coefficient map, of orders five and four; ℓ, m, s are its linear coordinate maps. Their rational entries are given in the complete proofs.

This lift attains the boundary. The positive definite Gram pair

$$G_0 = \mathrm{diag}(1,1,2,1,1), \quad J_0 = \mathrm{diag}(1,2,2,1)$$

represents $(\lambda, \alpha, h) = (1,0,0)$ and supplies strict feasibility. The normalization is $\mathrm{tr}(U(z)G_0) + \mathrm{tr}(V(z)J_0) = 1$, which bounds the PSD matrices and every auxiliary variable. The normalized dual image is compact and has exactly the support values in Eq. (5). It therefore equals $\mathcal{W}$, including its boundary.

Why the NPA obstruction remains. Equal overlaps determine support values; a finite NPA certificate must represent the bound for independently chosen measurement angles. The earlier nontermination proof [1] uses this freedom: near the local threshold, an NPA error bounded by a constant times the quantum violation above the local bound would force a shifted Motzkin polynomial to be a sum of polynomial squares. The univariate squares in Eq. (6) arise after the support minimization and need not be identities in a fixed list of observable words. The complete earlier argument is retained in the companion.

Minimal block order and thirty-two exact projections. For $f(\alpha) = \max_{\mathcal{W}}(S + \alpha M)$, the exact quantum bound has the expansion [1]

$$f(1-T) = 4 - 2T + \tfrac{4}{3}T^3 + O(T^4), \qquad T \downarrow 0. \quad (8)$$

A polygon would have affine support sufficiently close to $\alpha = 1$, contrary to Eq. (8). Thus $\mathcal{W}$ is nonpolyhedral. It is also closed, convex and semialgebraic, so Scheiderer's planar theorem gives $\mathrm{sxdeg}(\mathcal{W}) \leq 2$ [3]. Order-one lifts are polyhedral, hence $\mathrm{sxdeg}(\mathcal{W}) = 2$. The blocks $5, 4$ are explicit; the optimal-order lift is existential.

Physical input and output relabellings give the same quantum region for

$$\left(a_x + \chi_{xy} b_y,\ \sum_{ij} \chi_{ij} E_{ij} \right), \qquad \prod_{ij} \chi_{ij} = -1.$$

The eight CHSH sign patterns and four selected input pairs give 32 distinct planes. Intersecting their fixed lifts with any NPA relaxation makes every support value on these planes exact, including all the directions responsible for finite-level nontermination.

References

[1] A. Chaturvedi, *No Finite NPA Level Characterizes the Complete Quantum Set in the Simplest Bell Scenario*, manuscript (2026). Its complete nontermination proof is reproduced in the companion, Part II.

[2] C. Hanselka and R. Sinn, *Math. Z.* **292**, 83–101 (2019), Corollary 2.8; arXiv:1707.08261.

[3] C. Scheiderer, *SIAM J. Appl. Algebra Geom.* **5**, 114–139 (2021), Theorem 0.1; arXiv:2004.04196.